

Program: Diploma in Automation and Robotics	
Course Code: 6339B	Course Title: Advanced Microprocessor and Microcontroller Lab
Semester: 6	Credits: 1.5
Course Category: Program core	
Periods per week: 3 (L:0,T:0,P:3)	Periods per semester: 45

Course Objectives:

- To understand Assembly Language/embedded C programming of Microcontroller using trainer kit.
- To expertise working with Keil compiler and embedded C programming.
- Interface different input and output devices to PIC and control them using Assembly language programs/C Programs
- Interface different input and output devices to ARM

Course Prerequisites:

Topic	Course Code	Course Name	Semester
Basic concepts of computer programming.		Microcontroller based Robotic Application Design Lab	3

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize with the assembly level and embedded C programming using PIC.	18	Applying
CO2	Students will be able to have an in-depth knowledge on interfacing the external devices to the PIC	12	Applying
CO3	Students will be able to design a ARM based system with the help of the interfacing devices.	12	Applying
	Lab Exam	3	

CO–POMapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3				3		
CO2	3						
CO3	3			3			

3-Stronglymapped, 2-Moderatelymapped,1-Weaklymapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Familiarize with the assembly level and embedded C programming using PIC.		
M1.01	Addition, Subtraction, Multiplication and Division	3	Applying
M1.02	Largest/smallest from a series.	3	Applying
M1.03	Sorting (Ascending/Descending) of data.	6	Applying
M1.04	Data transfer/exchange between specified memory locations.	3	Applying
M1.05	Multiplication by shift and add method.	3	Applying
CO2	Students will be able to have an in-depth knowledge on interfacing the external devices to the PIC		
M2.01	Blinking LED using a switch	3	Applying
M2.02	Blinking LED with time delay	3	Applying
M2.03	8x2 and 16x2 Alphanumeric LCD interfacing	6	Applying
	Lab Exam	1.5	
CO3	Students will be able to design a ARM based system with the help of the interfacing devices.		

M3.01	ALP to multiply two 16 bit binary numbers	3	Applying
M3.02	ALP to find the sum of first 10 integer numbers	3	Applying
M3.03	Display “HELLO WORLD” message.	3	Applying
M3.04	Interface and control a DC motor	3	Applying
	LabExam II	1.5	

OnlineResources:

SINo	Website Link
1	https://nptel.ac.in/courses/117104072
2	https://nptel.ac.in/courses/106105193
3	https://www.microchip.com/en-us/product/PIC18F4550